Global Seismic Series: Statistical Analysis of Spatiotemporal Clustering in M≥6.5 Earthquake Catalogs, 1973–2026 CE

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Abstract.

We analyse an analysis catalog of 4,267 unique M≥6.5 events (modern window 1973–2026: 2,041 events; provenance: 4,418 CSV rows, ~151 NOAA M<6.5 excluded from clustering) using the Baiesi–Paczuski metric eta with tectonic-path distance (Bird 2003). The detector yields 47 algorithmic candidates (27 modern); historical NOAA records (n=47) reported in Appendix A only. Primary ETAS null (literature H&S 2003, decoupled): mean≈15.4, p_ETAS≤0.001 — deviation from ETAS indicates clustering the model does not fully explain; circum-Pacific; not global chains beyond model scope (Sec. 5.4). GK mainshocks: N=27 unchanged. Global-series hypothesis not supported (Sec. 5.4–5.6). Permutation p=0.0001 (1/10,001) rejects temporal Poisson null only. Limitations — Sec. 5.6.

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi–Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes do not occur as independent Poisson events. Following the 1992 Landers earthquake (M_w 7.3), Hill et al. [1993] documented remotely triggered seismicity at distances exceeding 1,000 km. Brodsky & Prejean [2006] showed that surface waves can initiate swarms in volcanic systems thousands of kilometres away. The 2004 Sumatra–Andaman earthquake (M_w 9.1) was followed by elevated activity in distant regions [Pollitz et al., 1998; Freed & Lin, 2001].

However, the systematic nature of such correlations remains debated. Michael [2011] found that M≥7 clustering in 1995–2011 is statistically indistinguishable from random fluctuations. Shearer & Stark [2012] reported no increase in global M≥7 and M≥8 rates after the 2004 Sumatra event. Kagan & Jackson [1999] confirmed elevated probability of paired events at short separation without resolving long-range links.

The ETAS model [Ogata, 1988] reproduces regional aftershock clustering but does not encode inter-plate correlations. The Baiesi–Paczuski [2004] metric and Zaliapin–Ben-Zion extensions [2008, 2013] provide objective cluster detection but typically use Euclidean distance, ignoring lithospheric connectivity.

Objective. Test (and if warranted, falsify) the hypothesis that physically meaningful multi-regional global series exist, with primary inference on the modern window (1973–2026), using complementary null tests (permutation vs ETAS) and explicit detector liberalness assessment. Scope. We combine nearest-neighbor clustering with heuristic tectonic hint and ETAS validation; this complements global rate tests (Michael 2011; Shearer & Stark 2012) with a different η-linkage statistic but does not supersede their conclusions.

2. DATA AND METHODS

2.1 Catalog compilation

Magnitude notation. Catalog thresholds and detector gates use M≥6.5 (USGS ComCat, predominantly M_w); individual events are cited with catalog type where relevant (e.g. M_w 7.3).

Three catalogs were merged: USGS ComCat (1900–2026, primary instrumental); ISC Bulletin (relocated hypocenters); and NOAA NGDC (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using ±30 days and ≤50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains 4,267 unique M≥6.5 events (from 4,418 CSV rows; ~151 M<6.5 excluded from clustering). USGS ComCat: 2,088 raw (1973–2026) → 2,041 after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). quality_score is metadata, not an inclusion filter. Primary analysis set: events 1900–2026 (4,218); 47 pre-1900 records remain in CSV (provenance) but are excluded from the primary detector pipeline and ETAS calibration window (1973–2026 only) — see Appendix A. Final catalog: 4,267 M≥6.5 events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b-value from 1,688 events above M_c:

b = 0.911 +/- 0.018 (n = 1,688 events)

2.2 Tectonic heuristic (deprecated)

The Bird (2003) tectonic-path heuristic is a deprecated diagnostic only (Appendix); primary analysis uses great-circle distance only.

eta_ij = t_ij * r_ij^1.6 * 10^(-b*mi)

b=1.0 — deliberate Baiesi & Paczuski (2004) simplification; catalog b=0.911±0.018 for M_c/completeness and MC null only — not in the η formula. Zaliapin (2008): D≈b — η, η_0, cluster shifts not tested.

2.3 Analysis pipeline

Raw catalogs → dedup → exclude M<6.5 (~151 NOAA) → GK declustering (primary) → eta NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: η NN + KDE threshold). At global M≥6.5 scale ZBZ is permissive (sparse events → high η). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary, N=27 likely unchanged (untested).

Why GK–ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low η at global catalog scale. Under fixed gates GK/ZBZ/none all yield N=27 (sensitivity_declustering.json); different mainshock labels (24 vs 1) could matter for cluster analysis.

2.4 Clustering and detector criteria

1. GK declustering (primary) → mainshocks for η NN forest.
2. η NN forest: b=1.0, $r^{1.6}$; tectonic Bird 2003 (1.5× GC fallback).
3. Sliding windows 1, 2, 5 yr (1-yr step); merge 142→47.
4. Detector gate: N≥4; M≥6.5; mean pairwise GC>1500 km.
5. Flinn–Engdahl zone count — diagnostic only (not admission criterion).

2.5 Primary ETAS null

Primary ETAS null uses literature H&S 2003 parameters, decoupled from detector output. Catalog calibration scripts are Appendix reproducibility only.

2.6 Statistical validation

Permutation test: n = 10,000, $p \leq 0.0001$, z = -6.17 (modern). ETAS validation (primary): literature H&S 2003: mean ≈ 15.4, $p_{ETAS} \leq 0.001$ — N_{obs} = 27 exceeds literature ETAS expectation; clustering not fully explained by model; circum-Pacific; not global chains. GK mainshocks only: N = 27 unchanged. BH post-hoc on N = 47 — not discovery.

3. RESULTS

3.1 Identified series

Full historical analysis yields 47 detector candidates (not validated global series): 27 modern ($p \leq 0.0001$), 15 early instrumental (p = 0.007; pre-1960 incompleteness caveat), 5 historical candidates (not significant, p = 0.46; 47 M≥6.5 events pre-1900). 142 cluster candidates before filtering.

Series ID	Events	Regions	Mmax	Years	Mean lat°	Mean lon°	Notes
1905–1910	193	43	8.8	1905–1910	−60…72	−180…180	Early instrumental; incomplete before ~1960
S047	53	5	8.0	1982–2024	−21.5…18.9	−175.5…155.2	Western Pacific
S170	46	12	9.1	2002–2023	−14.3…33.8	−113.1…167.3	Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	−8.1…16.5	120.4…146.4	Western Pacific arc
S116	22	5	8.2	1993–2021	−31.7…10.8	−179.5…179.4	South Pacific

Table 1. Top-5 detector candidates (not ETAS-validated physical series).

3.2 Spatial–temporal distribution

Elevated detector candidate frequency (not validated physical series) occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median $\Delta\log_{10}\eta$ = +0.28 on random pairs (mostly 1.5× GC fallback).

3.3 Consolidated sensitivity table (modern window)

Parameter	Setting	N_series
GC gate	1000 km	27
GC gate	1500 km (baseline)	27
GC gate	2000 km	27
Window	1 yr	53
Window	2 yr (baseline)	27
Window	5 yr	11

Parameter	Setting	N_series
Window	10 yr	6
b in η	1.0 (BP 2004)	27
b in η	0.911 (catalog)	27
Declustering	GK / ZBZ / none	27 / 27 / 27
min_events (strict)	5 / 6 / 8	27 / 27 / 27
Catalog	GK mainshocks only	27
η_0 ±20%	not_applied	—

Table 2. N_series sensitivity under fixed detector gates (mean GC > 1500 km, N ≥ 4). Sources: sensitivity_*.json.

4. DISCUSSION

The detector is liberal: under literature ETAS (mean ≈ 15.4, p_ETAS ≤ 0.001) N_obs exceeds local-clustering null. No physical mechanism; tectonic metric failed (98% GC fallback). Compatible with Michael (2011) and Shearer & Stark (2012).

5. CONCLUSIONS

The detector is liberal: literature ETAS yields mean ≈ 15.4, p_ETAS ≤ 0.001. Global-series hypothesis not supported (Sec. 5.4–5.6). Permutation rejects Poisson times only (Ogata 1988).

1. Heuristic with tectonic hint: 98% GC fallback — failed hypothesis test.
2. 47 detector candidates indistinguishable from ETAS null; ΔCFS/dynamic stress — future work only.

5.5 ETAS null limitations

Primary null uses literature H&S 2003 only; spatial Ogata (1998) MLE with confidence intervals is not implemented. Catalog calibration — Appendix B (reproducibility, not inference). The negative outcome also rests on no physical mechanism, failed tectonic metric (98% GC fallback), and liberal search (142 windows).

5.6 Limitations

Limitation	Affected step	Impact on main conclusion
η_0 unverified at global scale	GK declustering	N = 27 unchanged (global_series does not use η_0)
b = 1.0 vs 0.911 in η	η upstream	N_series = 27 both; cluster labels not re-run
No spatial Ogata MLE	ETAS null	Literature null only; not publication-grade catalog fit
142 windows + merge	Detector	Main source of liberalness
Literature p ≤ 0.001	ETAS test	ETAS-unexplained clustering; circum-Pacific; not global chains

Table 3. Limitation → affected step → impact on main conclusion (Sec. 5.6).

Impact analysis. N = 27 is computed by global_series() without η_0 filtering; b = 1.0 vs 0.911 leaves N_series = 27 under fixed gates, but upstream clusters at b = 0.911 were not re-verified. 142 sliding windows dominate detector liberalness.

APPENDIX A. PRE-1900 NOAA RECORDS

Forty-seven fragmentary paleoseismic/historical M≥6.5 records from NOAA NGDC are retained in data/processed/unified_catalog_full.csv for provenance (not removed). These 47 events are excluded from the primary detector pipeline and ETAS calibration window: pipeline_v2.py and calibrate_etas.py use the modern catalog 1973–2026 only (N=2,041). Epoch-stratified counts include pre-1900 descriptively via run_full_historical_analysis.py but do not enter primary significance claims.

APPENDIX B. CATALOG-MATCHED WLS NEGATIVE CONTROL

Reproducibility negative control only — not inference. Catalog-matched WLS (results/etas_calibration.json: μ≈0.103, K≈0.495) yields mean=27.0, p_ETAS=1.0 (n=1000; multiseed stable). Illustrates detector–calibration coupling; not used for inference. Do not cite p_ETAS=1.0 in abstract, conclusions, or hero metrics.

Component	Method
μ	GK mainshocks / T (closed form)
c, p	Omori MLE, Nelder–Mead on 24 delays
K, α	WLS (numpy.linalg.lstsq) on same 24 GK aftershocks

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md). External DOI deposition (Zenodo) deferred — GitHub only.

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